School of Mathematical Sciences
Dalian University of Technology
Prof. Xian Liao
Linggong Road 2
116024 Dalian, P.R.China
E-Mail: liao@dlut.edu.cn
Website: https://faculty.dlut.edu.cn/liao

Lecture Notes for "Fourier Analysis and Differential Equations" [1] [2]

Time [3] & Place:

Tuesday, 13:30-15:05, 研教楼 306
(Sep. 8, Sep. 15, Sep. 20, Sep. 22, Sep. 29, Oct. 13, Oct. 20, Oct. 27, Nov. 3)

Wednesday, 10:05-11:40, 综合教学 1 号楼 356
(Sep. 9, Sep. 16, Sep. 23, Sep. 30, Oct. 10, Oct. 14, Oct. 21, Oct. 28, Nov. 4)

Teaching assistant:
Dingli Jiang (dinglijiang1018(at)foxmail.com)

[1] The lecture notes are based on a lecture given at the Dalian University of Technology during the fall term 2026.
[2] Comments are welcome to be sent to me by email.
[3] It takes place from Sep. 6 to Nov. 6, 2026 (2.-10. semester week).

Contents

This two-month lecture provides an introduction to the sophisticated theory of Fourier analysis. The one-dimensional case is particularly emphasized, serving as motivation and providing examples for the illustration of the general theory. It also helps to connect the various concepts encountered in the theories of ordinary differential equations (ODEs), functional analysis, complex analysis, harmonic analysis and many other fields. Due to time constraints and the ambitious goal, we are unable to define all the related mathematical concepts rigorously, and instead aim to draw the audience's interest to the relevant theory. (The footnotes are left as exercises to the audience.) Finally the application to partial differential equations (PDEs) is demonstrated.

1 Fourier series

1.1 Examples of boundary value problems

1.1.1 An ODE with Dirichlet boundary condition

Consider the boundary value problem for a second-order ordinary differential equation on the interval $[0, \pi]$,

$$u''(x) + \lambda u(x) = 0 \text{ on } (0, \pi), \quad u(0) = u(\pi) = 0. \tag{1.1}$$

We consider three different cases:

- Case $\lambda < 0$.

  The following two functions are the fundamental solutions for the ODE $u''(x) = -\lambda u(x)$

  $$\sinh(\sqrt{-\lambda}x) = \frac{e^{\sqrt{-\lambda}x} - e^{-\sqrt{-\lambda}x}}{2},$$
  $$\cosh(\sqrt{-\lambda}x) = \frac{e^{\sqrt{-\lambda}x} + e^{-\sqrt{-\lambda}x}}{2},$$

  The boundary conditions imply that the solution should be zero. [4]

- Case $\lambda = 0$.

  A regular solution of $u''(x) = 0$ is a linear function $u(x) = ax + b$, which is 0 due to the boundary conditions.

[4] One can also argue from another perspective: We take the $L^2(0, \pi)$-inner product

- Case $\lambda > 0$.

  The following two functions are fundamental solutions for the ODE $u''(x) = -\lambda u(x)$

$$\sin(\sqrt{\lambda}x) = \frac{e^{i\sqrt{\lambda}x} - e^{-i\sqrt{\lambda}x}}{2i},$$
$$\cos(\sqrt{\lambda}x) = \frac{e^{i\sqrt{\lambda}x} + e^{-i\sqrt{\lambda}x}}{2},$$

  The boundary conditions imply that the solution should be a linear combination of

$$\sin(nx), \quad n = 1, 2, \cdots$$

To conclude, there is an infinite set of discrete values of λ:

$$\lambda = n^2, \quad n = 1, 2, \cdots$$

such that the boundary value problem (1.1) has a <u>nontrivial</u> solution. These values n^2, $n \in \mathbb{N}$ are called the eigenvalues of the (eigenvalue) problem (1.1) and the functions

$$u_n(x) = b_n \sin(nx), \quad n \in \mathbb{N}, \quad b_n \in \mathbb{R}(\text{or } \mathbb{C})$$

are the corresponding eigenfunctions. The concept of eigenvalues and eigenfunctions becomes clearer in Section 1.2 below.

1.1.2 Fourier's idea to solve the heat equation

Consider the one-dimensional initial-boundary value problem for the (real-valued) heat equation

$$\begin{cases} \partial_t u - \partial_{xx} u = 0 \text{ on } (0, \infty) \times (0, \pi), \\ u(0, x) = f(x) \quad x \in (0, \pi), \\ u(t, 0) = u(t, \pi) = 0 \quad t \in (0, \infty), \end{cases} \tag{1.3}$$

---

between the equation and the solution u itself, to obtain by integration by parts

$$-\int_0^\pi |u'|^2 \, \mathrm{d}x + \lambda \int_0^\pi |u|^2 \, \mathrm{d}x = 0. \tag{1.2}$$

This implies, if u is a regular solution (e.g. in $H^1(0, \pi)$), then $u = 0$. This argument can be easily generalized to the case when $\lambda \in \mathbb{C}$ with $\operatorname{Im} \lambda \neq 0$, where $u = 0$ (in $H^1(0, \pi)$) follows.

where the compatibility condition is satisfied

$$f(0) = f(\pi) = 0,$$

and we assume that f is not identically zero: $f \neq 0$.
J. Fourier[5] made the following Ansatz (separation of variables)

$$u(t,x) = T(t)X(x).$$

This leads to the following:

- $T'(t)X(x) - T(t)X''(x) = 0$, which implies (at least formally),

$$\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = -\lambda \text{ for some constant } \lambda.$$

- $T(t)X(0) = T(t)X(\pi) = 0$ for all time t, which implies

$$X(0) = X(\pi) = 0.$$

- $T(0)X(x) = f(x)$.

From the ODE results above, if $f \neq 0$ and one searches for nontrivial (real-valued) solution, λ can "only" take the value n^2 for some $n \in \mathbb{N}$, and $X(x)$ is a multiple of $\sin(nx)$, and in this case, the time function $T(t) = T(0)e^{-n^2 t}$. That is, if the initial data is

$$f(x) = b_n \sin(nx),$$

for some $n \in \mathbb{N}$ and some constant b_n, then the problem (1.3) has a (unique) solution

$$u(t,x) = T(t)X(x) = T(0)e^{-n^2 t}\frac{b_n \sin(nx)}{T(0)} = b_n e^{-n^2 t}\sin(nx).$$

If the initial data is a finite sum of sine functions

$$f(x) = \sum_{n=1}^{N} b_n \sin(nx),$$

then by the superposition principle the solution is

$$u(t,x) = \sum_{n=1}^{N} b_n e^{-n^2 t}\sin(nx).$$

[5] Joseph Fourier, *Théorie Analytique de la Chaleur.* Chez F. Didot, 1822 (Firstly presented to *Institute de France* in 1807).

Thus, formally, if we can expand the initial data $f(x)$ as a sine series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx), \tag{1.4}$$

then the following function (when it is well-defined) is a solution of (1.1)

$$u(x) = \sum_{n=1}^{\infty} b_n e^{-n^2 t} \sin(nx).$$

A function with such an expansion (1.4) (nowadays called Fourier sine series) "seems" very special. For half a century around 1800 there was a great debate whether such a series expansion holds only for specific functions.

1.2 Spectral theorem and Fourier series

We claim that any function in the functional space

$$\{f \in C^2([0, \pi]) \mid f(0) = f(\pi) = 0\} \tag{1.5}$$

can be expanded as Fourier sine series. Historically one can make use of the Weierstrass approximation theorem and the density of trigonometric functions in the space of polynomial functions to prove this statement. Here we illustrate this in modern mathematical language.
We can reformulate (1.1) as the following eigenvalue problem

$$\mathcal{A}u = \lambda u, \tag{1.6}$$

where the operator $\mathcal{A}$ is defined by[6]

$$\mathcal{A} : D(\mathcal{A}) \to X, \quad \mathcal{A}(u) = -u'' \text{ for } u \in D(\mathcal{A}). \tag{1.7}$$

Here we take the underlying inner product space and scalar product below

$$X = C([0, \pi]; \mathbb{C}), \quad \langle u, v \rangle = \int_0^\pi u\bar{v} \, \mathrm{d}x \,,$$

and the domain of the operator $\mathcal{A}$ is a subspace of X:

$$D(\mathcal{A}) = \{u \in C^2([0, \pi]) \mid u(0) = u(\pi) = 0\} \subset X.$$

[6] There is a certain conceptual mismatch between the classical setting of continuous functions and the spectral-theoretic framework, which is more naturally aligned with spaces such as $L^2(0, \pi)$. For example, one may define $\mathcal{A} : H_0^2(0, \pi) = D(\mathcal{A}) \to L^2(0, \pi)$.

The induced norm on X is

$$\|u\| = \langle u, u \rangle^{\frac{1}{2}},$$

and the distance between u and v is $\|u - v\|$. Note that $D(\mathcal{A})$ and X are not complete with respect to the defined metric.
Recall that the Lebesgue space

$$\begin{aligned} L^2(0,\pi) = &\{f \text{ is Lebesgue measurable on } (0,\pi) \mid \\ &\|f\|_{L^2(0,\pi)} = \Big(\int_0^\pi |f(x)|^2 \, \mathrm{d}x \Big)^{\frac{1}{2}} < \infty\} / \sim, \end{aligned}$$

where

- The integral is understood in the sense of Riemann-Lebesgue:

$$\|f\|_{L^2(0,\pi)} = \Big(2 \int_0^\infty \alpha \; m(\{x \in (0,\pi) \mid |f(x)| > \alpha\}) \, d\alpha \Big)^{\frac{1}{2}}$$

- The equivalence $\sim$ means identity almost everywhere (a.e.):

$$f \sim g \Leftrightarrow m(\{x \in (0,\pi) \mid f(x) \neq g(x)\}) = 0.$$

The Lebesgue space $L^2(0,\pi)$ is endowed with the scalar product $\langle \cdot, \cdot \rangle$, and is a Hilbert space. By the density of the space[7] of test functions $C_c^\infty(0,\pi)$ in $L^2(0,\pi)$, the closure of $C_c^\infty(0,\pi) \subset D(A) \subset X$ in the Lebesgue space $(L^2(0,\pi), \langle \cdot, \cdot \rangle)$ is the Lebesgue space itself

$$\overline{X}^{L^2} = \overline{D(\mathcal{A})}^{L^2} = \overline{C_c^\infty(0,\pi)}^{L^2} = L^2(0,\pi).$$

1.2.1 Resolvent computation

From the ODE results above (recalling (1.2)), the eigenvalues and normalized eigenfunctions of the operator $\mathcal{A}$ are

$$\lambda_n = n^2, \quad u_n = \sqrt{\frac{2}{\pi}} \sin(nx), \quad n \in \mathbb{N}, \tag{1.8}$$

with $\|u_n\|_{L^2(0,\pi)} = 1$.

[Sep 8, 2026]

[Sep 9, 2026]

[7]The density result also holds on the whole line or in higher dimensions.

We now compute the resolvent of $\mathcal{A}$ at $\lambda = -1 \notin \mathbb{N}$, that is, we compute $R_{\mathcal{A}}(-1) = (\mathcal{A}+1)^{-1} : X \to D(\mathcal{A})$. That is, for a given function $g \in X$, we seek the solution $u \in D(\mathcal{A})$ of the following boundary value problem for the inhomogeneous ODE:

$$-u'' + u = g, \quad u(0) = u(\pi) = 0. \tag{1.9}$$

It suffices to compute the Green function of this boundary value problem. The homogeneous equation $-u'' + u = 0$ has fundamental solutions

$$e^{-x}, \quad e^{x}.$$

The solution satisfying the boundary condition $u(0) = 0$ is a multiple of

$$\phi(x) = e^{x} - e^{-x},$$

while the solution satisfying the boundary condition $u(\pi) = 0$ is a multiple of

$$\psi(x) = e^{-(\pi - x)} - e^{\pi - x}.$$

Their Wronskian $W(\phi,\psi) = \phi\psi' - \phi'\psi = \det\begin{pmatrix} \phi & \psi \\ \phi' & \psi' \end{pmatrix}$ is independent of x [8] and is given by

$$W(\phi,\psi) = 2(e^{\pi} - e^{-\pi}).$$

The Green function is

$$\begin{aligned} G(x,y) &= \begin{cases} \frac{\phi(x)\psi(y)}{W(\phi,\psi)} & \text{for } x < y \\[2ex] \frac{\phi(y)\psi(x)}{W(\phi,\psi)} & \text{for } x > y \end{cases} \\ &= \frac{e^{-\pi+x+y} + e^{\pi-(x+y)} - e^{\pi-|x-y|} - e^{-\pi+|x-y|}}{W(\phi,\psi)}. \end{aligned}$$

Thus the unique solution of (1.9) reads[9]

$$u(x) = \int_0^{\pi} G(x,y) g(y) \, \mathrm{d}y \,, \tag{1.10}$$

[8] It follows from Abel's identity (also called Liouville's formular).

[9] The unique solvability of the inhomogeneous problem (1.9) follows from the fact that the homogeneous problem (1.1) with $\lambda = -1$ has only trivial zero solution, by the Fredholm alternative theorem. The unique solution of (1.9) can be represented by Green function (1.10): $u(x) = \int_0^x G(x,y)g(y)\,\mathrm{d}y + \int_x^{\pi} G(x,y)g(y)\,\mathrm{d}y$, where the Green function has the following properties

- $G(\cdot, y)$ satisfies the homogeneous ODE (with respect to x variable) and the boundary conditions;
- $G(x,y) \in C([0,\pi]^2)$, $G(x,y) \in C^2(\Delta_{\pm})$ with $\Delta_{\pm} = \{(x,y) \in [0,\pi]^2 \mid \pm(x-y) \leqslant 0\}$, and $\partial_x G$ has a jump at the diagonal line: $\partial_x G(x, x-0) - \partial_x G(x, x+0) = 1$.

that is,

$$R_{\mathcal{A}}(-1)g(x) = (\mathcal{A}+1)^{-1}g(x) = \int_0^\pi G(x,y)g(y)\,\mathrm{d}y\,.$$

One may verify that

- $R_{\mathcal{A}}(-1) : X \to D(\mathcal{A})$.

  This follows immediately from (1.9).

- $R_{\mathcal{A}}(-1)$ is a compact operator in X. That is, for any bounded sequence $(g_k)_k$ in X, the function sequence $(R_{\mathcal{A}}(-1)g_k)_k$ has a subsequence which converges in X.

  This follows from the properties of G and a compactness argument, e.g. Arzela-Ascoli theorem.

- $R_{\mathcal{A}}(-1)$ is symmetric. That is,

  $$\big\langle R_{\mathcal{A}}(-1)g, h\big\rangle = \big\langle g, R_{\mathcal{A}}(-1)h\big\rangle.$$

  This follows also from the symmetry property of $G(x,y)$.

1.2.2 Spectral theorem and Fourier sine series

Spectral theorem[10] states that given an infinte-dimensional inner product space X and a compact symmetric operator $\mathcal{R} : X \to X$, there exists a sequence of real eigenvalues $(z_n)_n \subset \mathbb{R}$ converging to 0. The corresponding normalized eigenvectors $(v_n)_n$ form an orthonormal set and every $f \in \mathrm{Ran}(\mathcal{R})$ can be written as

$$f = \sum_{n=1}^{\infty}\langle f, v_n\rangle v_n.$$

If $\mathrm{Ran}(\mathcal{R})$ is dense, then the eigenvectors form an orthonormal basis.
Thus for the compact symmetric operator $R_{\mathcal{A}}(-1)$ in X, there exists a sequence of real eigenvalues $(z_n) \to 0$ and the corresponding normalized eigenfunctions

$$(v_n) \subset D(\mathcal{A}), \quad \|v_n\| = \langle v_n, v_n\rangle = 1,$$

which form an orthonormal basis in the closure of its range $\overline{\mathrm{Ran}(R_{\mathcal{A}}(-1))}^{L^2} = \overline{D(\mathcal{A})}^{L^2} = L^2(0,\pi)$. That is, for any $f \in L^2(0,\pi)$, we can expand it as follows:

$$f = \sum_{n=1}^{\infty}\langle f, v_n\rangle v_n.$$

[10] We don't give the proof here.

Now we return to the operator $\mathcal{A}$. The fact that $R_{\mathcal{A}}(-1)v_n = z_n v_n$ is equivalent to

$$\mathcal{A}v_n = (-1 + \frac{1}{z_n})v_n,$$

that is, v_n is a normalized eigenfunction of $\mathcal{A}$ associated with the eigenvalue $-1 + \frac{1}{z_n}$ which tends to infinity. Recalling (1.8), we find

$$\lambda_n = n^2 = -1 + \frac{1}{z_n}, \quad u_n = \sqrt{\frac{2}{\pi}}\sin(nx) = v_n.$$

To conclude, the expansion $f = \sum_{n=1}^{\infty}\langle f, v_n\rangle v_n$ for any function $f \in L^2(0,\pi)$ is indeed (1.4):

$$f = \sum_{n=1}^{\infty} b_n \sin(nx), \quad b_n = \sqrt{\frac{2}{\pi}}\langle f, v_n\rangle = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)\,\mathrm{d}x\,.$$

This is the Fourier sine series. Let us emphasize here that the series is converges with respect to our scalar product, i.e. in the sense of $L^2(0,\pi)$. We remark that if the function $f \in D(\mathcal{A})$, then the sine series converges uniformly to f. The convergence of Fourier series is discussed in more detail in Section 1.3 later.

1.2.3 Eigenvalue problem with Neumann boundary condition

Instead of Dirichlet boundary condition studied in Section 1.2, we consider the Laplace eigenvalue problem associated to the Neumann boundary condition. More precisely, we consider the eigenvalue problem

$$\mathcal{B}u = \lambda u, \quad \mathcal{B} : D(\mathcal{B}) \to X, \quad \mathcal{B}(u) = -u'' \text{ for } u \in D(\mathcal{B}), \tag{1.11}$$

where the domain of the operator $\mathcal{B}$ is

$$D(\mathcal{B}) = \{u \in C^2([0,\pi]) \,|\, u'(0) = u'(\pi) = 0\} \subset X,$$

with $X = C([0,\pi];\mathbb{C})$ equipped with the inner product $\langle\cdot,\cdot\rangle$. The same analysis as before yields the following eigenvalues and normalized eigenfunctions

$$\lambda_0 = 0, \quad \lambda_n = n^2, \quad u_0 = \sqrt{\frac{1}{\pi}}, \quad u_n = \sqrt{\frac{2}{\pi}}\cos(nx), \quad n \in \mathbb{N}\,.$$

We can then compute the resolvent $R_{\mathcal{B}}(-1) : X \to D(\mathcal{B})$ and verify that it is compact and symmetric. Finally the spectral theorem ensures that any function $f \in L^2(0,\pi)$ can be expanded as Fourier cosine series

$$f = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx),$$
$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,\mathrm{d}x\,, \quad a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\,\mathrm{d}x\,, \quad n \in \mathbb{N},$$

in the sense of $L^2(0,\pi)$.

1.2.4 Full Fourier series

We now consider the eigenvalue problem

$$\mathcal{C}u = \lambda u, \quad \mathcal{C} : D(\mathcal{C}) \to Y, \quad \mathcal{C}(u) = -u'' \text{ for } u \in D(\mathcal{C}), \tag{1.12}$$

where

- $Y = C([-\pi,\pi];\mathbb{C})$ is equipped with the inner product $\langle\cdot,\cdot\rangle_{L^2(-\pi,\pi)}$;
- we assume periodic boundary conditions on the interval $[-\pi,\pi]$

$$D(\mathcal{C}) = \{u \in C^2([-\pi,\pi]) \mid u(-\pi) = u(\pi), \quad u'(-\pi) = u'(\pi)\} \subset Y.$$

There are eigenvalues

$$\lambda_0 = 0, \quad \lambda_n = n^2, \quad n \in \mathbb{N},$$

and the corresponding normalized eigenfunctions

$$u_0 = \sqrt{\frac{1}{2\pi}}, \quad \{u_n, v_n\} = \left\{\sqrt{\frac{1}{\pi}}\cos(nx),\ \sqrt{\frac{1}{\pi}}\sin(nx)\right\}, \quad n \in \mathbb{N}\,.$$

The spectral theorem implies that any function $f \in L^2(-\pi,\pi)$ can be expanded as Fourier series in the sense of $L^2(-\pi,\pi)$:

$$f = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx), \quad a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,\mathrm{d}x\,,$$
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,\mathrm{d}x\,, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,\mathrm{d}x\,, \quad n \in N.$$

Observe that

- If $f \in C([-\pi,\pi])$ is an odd function, then $a_0 = a_n = 0$, and hence the full Fourier series of any odd function on $(-\pi,\pi)$ coincides with Fourier sine series on $(0,\pi)$ in the sense of $L^2(0,\pi)$.

  Conversely, if $f \in C([0,\pi])$ with Dirichlet boundary condition $f(0) = f(\pi) = 0$, then one can extend it to an odd function in $C([-\pi,\pi])$ satisfying $f(-\pi) = f(0) = f(\pi) = 0$, and the two Fourier series coincide on $(0,\pi)$ in the sense of $L^2(0,\pi)$.

  This is the physical intuition why the Fourier sine series is "complete" on $[0,\pi]$.

- If $f \in C([-\pi,\pi])$ is an even function, then $b_n = 0$, and hence the full Fourier series of any even function on $(-\pi,\pi)$ coincides with Fourier cosine series on $(0,\pi)$ in the sense of $L^2(0,\pi)$.

  Conversely, if $f \in C([0,\pi])$ with Neumann boundary condition $f'(0) = f'(\pi) = 0$, then one can extend it to an even function in $C([-\pi,\pi])$ satisfying $f'(-\pi) = f'(0) = f'(\pi) = 0$, and the two Fourier series coincide on $(0,\pi)$ in the sense of $L^2(0,\pi)$.

- Replacing

  $$\cos(nx) = \frac{1}{2}(e^{inx} + e^{-inx}), \quad \sin(nx) = \frac{1}{2i}(e^{inx} - e^{-inx}),$$

  we can rewrite the full Fourier series in the elegant complex form

  $$\begin{aligned}
  &f = \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad \frac{1}{\sqrt{2\pi}} c_0 = a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-i0x} \, \mathrm{d}x \,, \\
  &\frac{1}{\sqrt{2\pi}} c_n = \frac{1}{2} a_n + \frac{1}{2i} b_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} \, \mathrm{d}x \,, \quad n \in \mathbb{N}, \\
  &\frac{1}{\sqrt{2\pi}} c_{-n} = \frac{1}{2} a_n - \frac{1}{2i} b_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-i(-n)x} \, \mathrm{d}x \,, \quad n \in \mathbb{N} \,.
  \end{aligned}$$

References

[1] H. Bahouri, J.-Y. Chemin and R. Danchin: Fourier analysis and nonlinear partial differential equations. Springer, 2011.

[2] J. Duoandikoetxea: Fourier Analysis. Graduate Studies in Mathematics, Vol. 29. American Mathematical Society, 2001.

[3] L. Grafakos: Classical Fourier Analysis. Springer, 2014.

[4] M. Haragus, M.A. Johnson, W.R. Perkins, Linear Modulational and Subharmonic Dynamics of Spectrally Stable Lugiato-Lefever Periodic Waves, *J. Differential Equations*, 280 (2021), 315-354.

[5] G. Teschl, Ordinary Differential Equations and Dynamical Systems, Version: July 22, 2009.